

Solution Requirements

Date	8 July 2026
Team ID	SWTID-2026-5447
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviours, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system shall provide secure login using username and OTP for Admin to access applicant data.	High
2	Authorization levels	<i>Admin: View reports and prediction history.</i> Applicant: Submit application details and view approval status.	High

3	External interfaces	No external API integration. Uses user input and ML for prediction.	High
4	Transactions processing	The system shall process the credit card application, run the ML prediction, and store the result as Approved or Rejected.	High
5	Reporting	<i>The system shall generate reports and dashboards showing total applications, approval rate, and rejected applications for Admin.</i>	Medium
6	Business rules, etc.	Business Rule: The application approval depends on ML prediction based on income, employment, credit score, and other features.	High
7	Compliance with laws or regulations	The system shall comply with RBI guidelines and data privacy standards, such as GDPR, to protect applicant personal and financial data.	High
8	<i>Other</i>	The system shall allow Admin to retrain the ML model with new applicant data to improve	Medium

Step 2: Non-Functional Requirements (NFR)

S No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system shall provide prediction results quickly after form submission.</i>	<i>Prediction results in less than 5 seconds</i>
2	Scalability	The system shall handle an increasing number of applicants and data without performance issues.	Support 1000+ concurrent applications
3	Security & Data Privacy	<i>All applicant personal and financial data shall be encrypted and access shall be restricted to authorized Admin only.</i>	<i>AES-256 encryption and role-based access control</i>
4	Reliability & Availability	The system shall be available to applicants and Admin with minimal downtime.	99.5% uptime per month
5	Usability & Accessibility	<i>The application interface shall be easy to use and accessible on desktop and mobile devices.</i>	Responsive UI, WCAG 2.1 AA compliant.

6	Other	<i>The system shall be accessible from multiple browsers and devices without installation.</i>	Compatible with Chrome, Firefox, Edge, Android and iOS
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